

Homework 3: Probability and Boxplots

Name:

Class Start Time:

The homework is broken into three parts. One will be on basic probability; one will be on the normal distribution; and one will be a “stretch” goal.

1 General Probability

We will assume we have a bag full of marbles and cubes. Assume we have 100 object in total in the bag with the following counts

Color	Marble	Cube
Red	10	20
Blue	10	30
Green	20	10

1. What is the probability of drawing a cube?

Ie what is $P(X = \text{cube})$? where X is the result of a single random draw.

$$\frac{60}{100} = .6$$

2. What is the probability of drawing something green AND it being a cube?

Ie What is $P(X = \text{green} \cap \text{cube})$?

$$\frac{10}{100} = .1$$

3. What is the value for $P(X = \text{green})$?

$$\frac{30}{100} = .3$$

4. What is the probability of drawing a green marble, given that we have drawn a marble?

Ie What is $P(X=\text{green} \cap \text{marble} | \text{marble}) = P(X=\text{green} | \text{marble})$?

$$\frac{20}{40} = .5$$

NOTE: We have a new denominator because we are only looking at the marble subpopulation

5. What is $P(X = \text{Blue} \cup \text{Cube})$? That is, what is the probability of drawing something that is either blue or a cube?

$$\frac{10+30+20+10}{100} = .7$$

NOTE: We want everything that is either blue, or a cube, or both

6. What is $P(X=\text{marble})$?

$$\frac{40}{100} = .4$$

7. What is the $P(X=\text{red})$?

$$\frac{30}{100} = .3$$

8. What is the $P(X = \text{red marble} | \text{red})$?

$$\frac{10}{30} = \frac{1}{3}$$

9. What is $P(X = \text{red marble} | \text{marble})$?

$$\frac{10}{40} = \frac{1}{4}$$

10. Multiple your answers to questions 8 and 7, then divide by your answer to question 6. How does this compare to question 9? (This is called Bayes' Theorem)

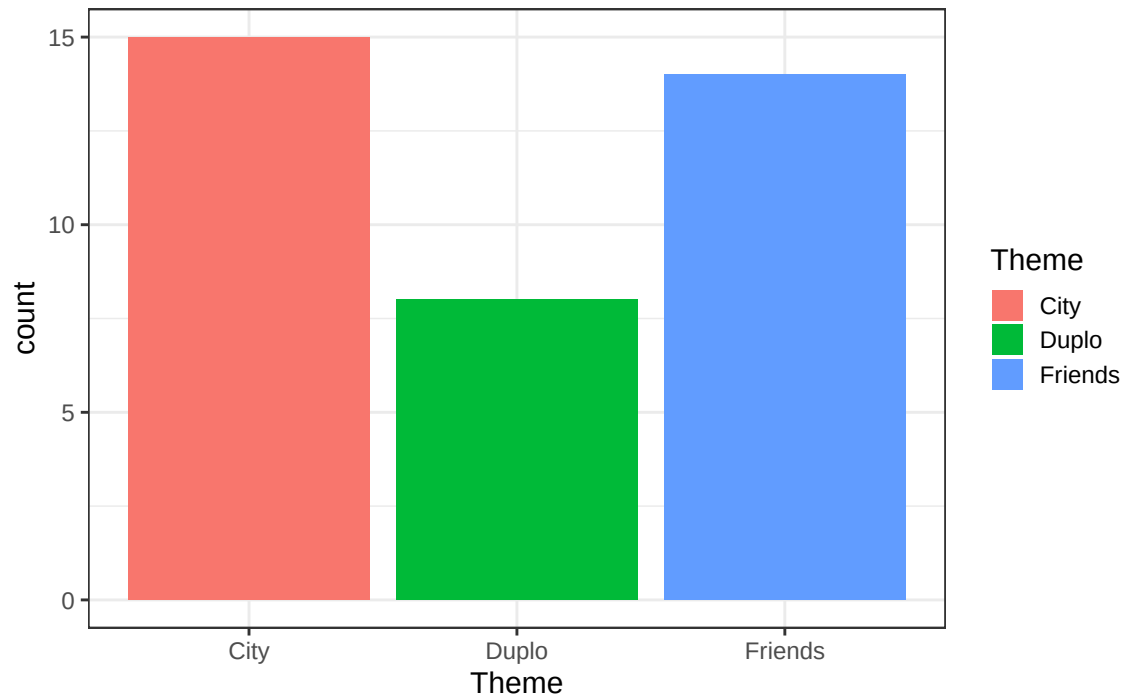
$$\frac{\frac{3}{10} * \frac{1}{3}}{(\frac{4}{10})} = \frac{4}{10}$$

Which is the same number as question 9.

2 Barplots and Probability

Let's return to the lego sets for a moment. Below is the barplot for the three themes of lego sets we studied (Duplo, Friends, City).

NOTE: I removed all lego sets that cost under \$25. That's not relevant to the problem but it's why the data here doesn't line up perfectly with the last homework.



1. Please describe the above barplot. Mention the largest and smallest categories and any patterns that seem important.

City is the highest category with 15 but Friends is close with 14. Duplo has the least with only 8.

2. Can the order of the three bars be rearranged? If so, why? If not, why not?

Yes they can; it's because the variables are nominal categories so the ordering doesn't matter.

3. Approximately how many lego sets are in....

4. You'll need your answer to question 3 for this. Assume Statistics Claus (Santa Claus's data-junky cousin) has taken all the above lego sets and put them in his magical bag. You've been a good (enough) student so he allows you to draw out a single lego set at random.

- (a) Approximately what is the probability you'd draw a City lego set?

In total I have 37 lego sets I believe so....

City?	Duplo?	Friends?
15	8	14

(b) Approximately what is the probability you'd draw a Friends lego set?

$$\frac{14}{37}$$

(c) Approximately what is the probability you'd draw a Duplo lego set? $\frac{8}{37}$

(d) Approximately what is the probability you draw a Friends or City lego set?

$$\frac{14+15}{37} = \frac{29}{37}$$

(e) Approximately what is the probability you don't draw a Duplo lego set?

$$\frac{14+15}{37} = \frac{29}{37} = 1 - \frac{8}{37}$$

(f) What is the probability you draw a Dinosaur lego set? (HINT: There are only Friends, City, and Duplo lego sets in Santa's bag)

$$\frac{0}{37} = 0$$

5. Please rewrite the six questions above using probability notation that was used in class as well as the first two pages.

(a) $P(\text{City})$

(b) $P(\text{Friends})$

(c) $P(\text{Duplo})$

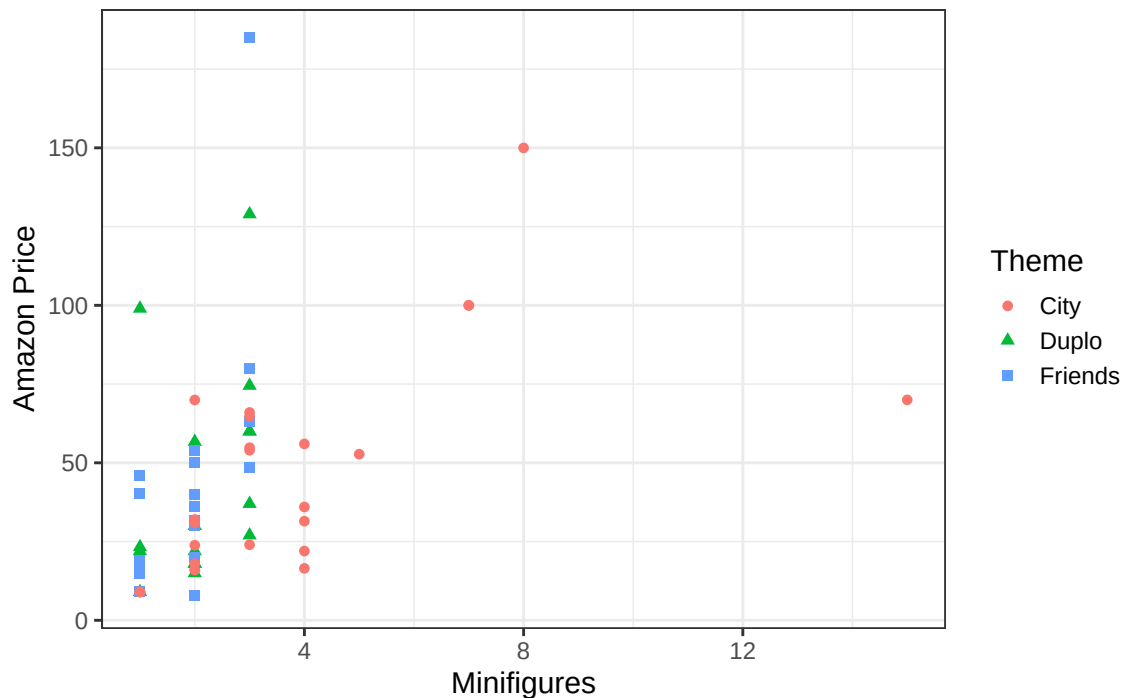
(d) $P(\text{Friends} \cup \text{City})$

(e) $P(\text{Duplo}^c)$

(f) $P(\text{Dino})$

3 Scatterplots Reviewed

Below is the scatterplot for the number of minifigures in the lego set and the price on Amazon.



1. Please describe the form, direction, outliers, and strength for the above graph. Ignore the different themes for the moment.

Very weak but generally positive. Yes, there is at least one major outlier in the x-direction. For form I'd waffle between a poorly fitting line and just saying a cloud of points.

2. There seems to be a major outlier in the city theme'd lego sets. After digging into the data file the lego set is called "People Pack - Outdoor Adventures". Comment on why this lego set is an outlier and whether it seems like it belongs with the other lego sets. You may want to google this lego set to learn more about it.

The lego set seems like an outlier in that it doesn't let you "build" anything but instead is an accessory for other lego sets. Because of that it seems like it doesn't fit with the other data points (generally you buy legos to actually *build* something).

3. When I've been making this scatterplots I have used both color (red, green, and blue) and shape (circle, triangles, and squares). The process of marking a variable on a graph in multiple ways is called "redundant encoding". Why do you think this is a good idea? Hint: Think about colorblind people....or greyscale printing

Redundant encoding allows for accessibility for colorblind people. Red, green, and blue circles could look similar. The same effect happens when printed in grey scale. Shapes add a secondary way to read the "Theme" variable that doesn't depend on the readability of color.

NOTE: Color should still generally be included as it allows for rapid division of the graph for poeple with full color spectrum sight. If we used just black shapes *everyone* would have to spend more time reading the graph to break it into it's respective subgroups.